

CHAPTER - III

SCOPING

3.1 :::: Introduction:

- i). Scope refers to the accessibility of the variable and functions in one part of the program to another part of the program.
- ii). Generally a variable declared in a program can receive global scope.
- iii). Global Scope variable can be used throughout the program.

3.2 :::: Variable Scope:

Mapping:

- i). The process of binding a variable with an object is called as Mapping.
- ii). $:=$ (equal sign) is used to map variable with an object.
- iii). Programming languages keeps track of mappings with namespace.
- iv). Namespaces are containers for mapping names of variables to objects.



A **namespace** is a simple system to control the names in a program. It ensures that names are unique and won't lead to any conflict. Also, add to your knowledge that **Python** implements **namespaces** in the form of dictionaries. It maintains a name-to-object mapping where names act as keys and the objects as values.

[www.techbeamers.com > python-namespace-scope](http://www.techbeamers.com/python-namespace-scope)

Python Namespace and Scope - Understand with Examples

a). Example:

- a:=5 where the integer value 5 is mapped to variable (namespace) a.
- b:=10 where the integer value 10 is mapped to variable (namespace) b.

Scope of Variable:

- i/. Scope refers to the accessibility of the variable and functions in one part of the program to another part of the program.
- ii/. Scope of a variable refers to part of the code where variable is visible.
- iii/. Example:

```
Disp( ):
```

```
    a:=7
```

- iv/. When user try to access 'a' outside "Disp()" function then "name 'a' is not defined" error is displayed. Because the lifetime of variable is only within the function "Disp()".
- v/. The duration for which a variable is alive is called as "Life Time"

3.3 :::: LEGB Rule:

- i/. Scope defines the order in which variables have to be mapped to the object in order to obtain the value.
- ii/. Example:

```
x:="Outer X Variable"
print x
display():
    x:="Inner X Variable"
    print x
display( )
```

Output:

Outer X Variable

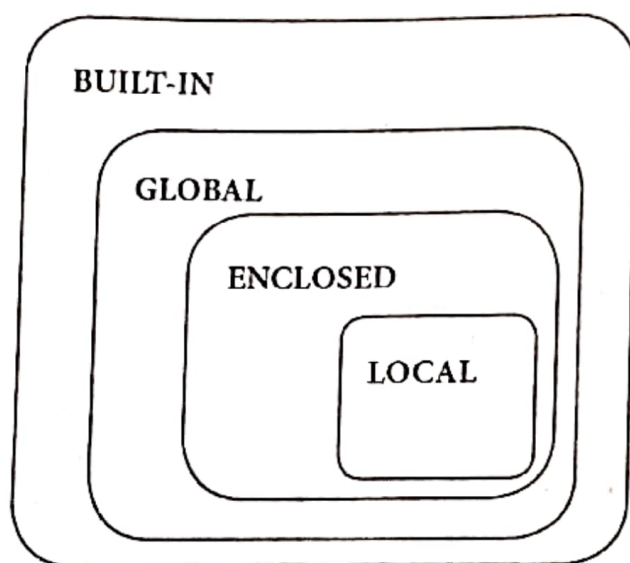
Inner X Variable

LEGB Rule:

- i/. LEGB rule is used to decide the order in which the scopes are to be searched.
- ii/. The following list shows the various scopes in terms of hierarchy.

- a. Local (L)
- b. Enclosed (E)
- c. Global (G)
- d. Built in (B)

Local(L)	Defined inside function/class
Enclosed(E)	Defined inside enclosing functions (Nested function concept)
Global(G)	Defined at the uppermost level
Built-in (B)	Reserved names in built-in functions (modules)



3.4 :::: Types of Variable Scope:

There are four types of scopes are available in Python. They are,

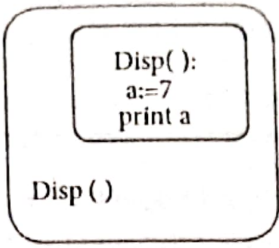
- i). Local
- ii). Enclosed
- iii). Global
- iv). Built-in

Local Scope:

- i). Local Scope refers to variables defined in current function.
- ii). Always Function will first look up the variable in Local Scope.

iii). If variable is not in Local scope then function will look up in outer scopes.

iv). Example:

1. Disp():	Entire program	Output of the Program
2. a:=7		7
3. print a		
4. Disp()		

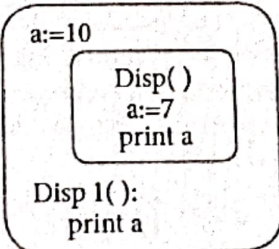
o). The above code displays the variable 'a' value as 7. Because 'a' is defined and available in Local Scope.

Global Scope:

i). Global Scope refers to a variable defined in outside of all function in a program.

ii). Global scope variable can be accessed anywhere of the program.

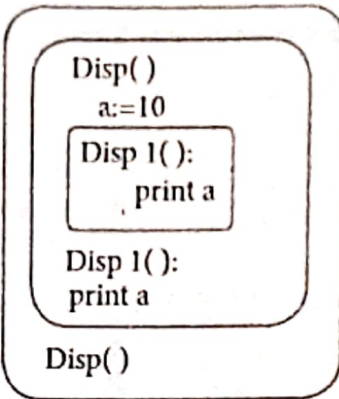
iii). Example:

1. a:=10	Entire program	Output of the Program
2. Disp():		7
3. a:=7		10
4. print a		
5. Disp()		
6. print a		

iv). The above code displays the variable 'a' value as 7 (local scope value) and then 10 (global scope value).

Enclosed Scope:

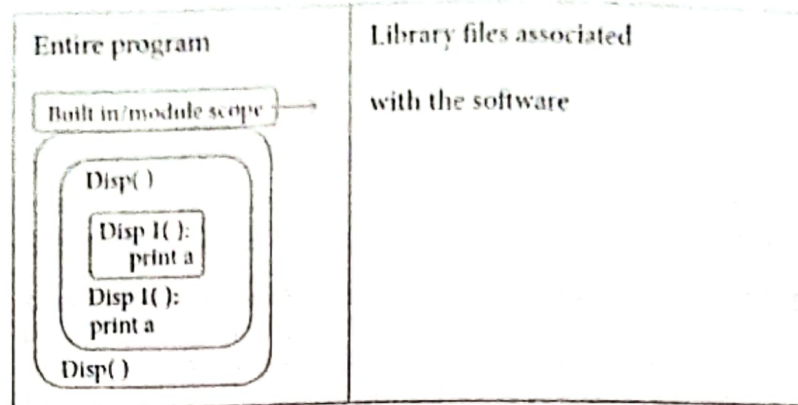
- i). Enclosed Scope refers to a variable defined inside the nested function in a program.
- ii). The inner function can access the variables declared in outer function. This is called as Enclosed scope.
- iii). Example:

1. Disp():	Entire program	Output of the Program
2. a:=10		10
3. Disp1():		10
4. print a		
5. Disp1()		
6. print a		
7. Disp()		

- iv). The above code displays the variable 'a' value as 10 (enclosed scope value) and then 10 (local scope value).

Built-in Scope:

- i). Built-in Scope refers to a variable defined in library function of a programming language.
- ii). Built-in Scope variables are loaded when library files are imported into a program.



3.5 :::: Module:

Module:

Program is comprises of one or more independently developed modules. Modules works perfectly. Modules can integrate with other modules. Using modules, programming and debugging is easily done.

Modular Programming:

A program can be divided into small functional modules which work together to get the output. The process of subdividing a program into several sub programs is called as Modular Programming.

Modular programming enables the programmer to divide the program and debug the program very easily. The following list show the example for modules.

- Procedure
- Subroutines
- Functions

Characteristics of Modules:

The following are the characteristics of modules

- i). Module can be included in a program.
- ii). Module contains instructions, logic and data.
- iii). Module can be separately compiled and stored in library.
- iv). Module segments can be invoked by using name and arguments.

n). Module segments can be used in another module if required.

Merits of Modular Programming:

The following are the merits of modular programming.

- i). Less code to be written
- ii). Modules can be reused.
- iii). Modular Programs can be easily developed.
- iv). Module code is short, simple and easy to understand.
- v). In modular programming, debug is very easy.
- vi). The module code is stored across multiple files.
- vii). Module eliminates the retype the code many times in an application.
- viii). The same module can be used in different application.
- ix). The scope of variable is easily controlled.
- x). Modular programming enables more programmers work on same application.

Access Control:

- i). Access control is a security technique.
 - ii). Access control regulates who can view or use resources in computing environment.
 - iii). Access control is a fundamental concept in security.
 - iv). Access control which minimize the risk of an object.
 - v). In other words, Access control is a selective restricted access of data.
 - vi). In OOPS (C++ & Java), Access control is done by Access Modifiers such as
 - a. Private
 - b. Protected
 - c. Public
- Private members can be accessed only within the class.
 - Protected members can be accessed within the class (Parent class) and its derived class (Child class).
 - Public members can be accessed outside the class.

- By default, all members in a python class are Public.
- By default, all members in a C++ and Java are Private.
- Any member can be accessed from outside the class environment in python but not possible in C++ and Java.
- In Python,
 - A variable with single underscore is protected variable.
 - A variable with double underscore is private variable.

Data Encapsulation:

The arrangement of private variable and public function ensures the principle of data encapsulation.

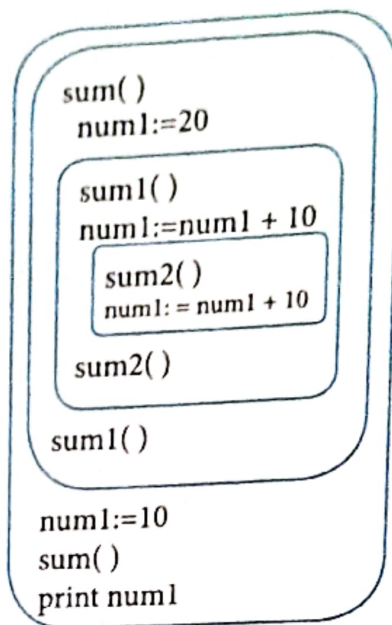
Points to remember:

- Scope refers to the visibility of variables, parameters and functions in one part of a program to another part of the same program.
- The process of binding a variable name with an object is called mapping. = (equal to sign) is used in programming languages to map the variable and object.
- Namespaces are containers for mapping names of variables to objects.
- The scope of a variable is that part of the code where it is visible.
- The LEGB rule is used to decide the order in which the scopes are to be searched for scope resolution.
- Local scope refers to variables defined in current function.
- A variable which is declared outside of all the functions in a program is known as global variable.
- A function (method) within another function is called nested function.
- A variable which is declared inside a function which contains another function definition within it, the inner function can also access the variable of the outer function. This scope is called enclosed scope.
- Built-in scope has all the names that are pre-loaded into program scope when we start the compiler or interpreter.
- A module is a part of a program. Programs are composed of one or more independently developed modules.
- The process of subdividing a computer program into separate sub-programs is called Modular programming.
- Access control is a security technique that regulates who or what can view or use resources in a computing environment. It is a fundamental concept in security that minimizes risk to the object.
- Public members (generally methods declared in a class) are accessible from outside the class.
- Protected members of a class are accessible from within the class and are also available to its sub-classes.
- Private members of a class are denied access from the outside the class. They can be handled only from within the class.
- Python prescribes a convention of prefixing the name of the variable/method with single or double underscore to emulate the behaviour of protected and private access specifiers.
- C++ and Java, control the access to class members by public, private and protected keywords.
- All members in a Python class are public by default whereas by default in C++ and Java all members are private.



Hands on Practice

1. Observe the following diagram and Write the pseudo code for the following



Evaluation

Part - I

Choose the best answer

(1 Mar

- Which of the following refers to the visibility of variables in one part of a program and another part of the same program.
 - Scope
 - Memory
 - Address
 - Accessibility
- The process of binding a variable name with an object is called
 - Scope
 - Mapping
 - late binding
 - early binding
- Which of the following is used in programming languages to map the variable and object
 - ::
 - :=
 - =
 - ==
- Containers for mapping names of variables to objects is called
 - Scope
 - Mapping
 - Binding
 - Namespaces
- Which scope refers to variables defined in current function?
 - Local Scope
 - Global scope
 - Module scope
 - Function Scope

6. The process of subdividing a computer program into separate sub-programs is called

(A) Procedural Programming	☒ (B) Modular programming
(C) Event Driven Programming	(D) Object oriented Programming
7. Which of the following security technique that regulates who can use resources in a computing environment?

(A) Password	(B) Authentication
☒ (C) Access control	(D) Certification
8. Which of the following members of a class can be handled only from within the class?

(A) Public members	(B) Protected members
(C) Secured members	☒ (D) Private members
9. Which members are accessible from outside the class?

☒ (A) Public members	(B) Protected members
(C) Secured members	(D) Private members
10. The members that are accessible from within the class and are also available to its sub-classes is called

(A) Public members	☒ (B) Protected members
(C) Secured members	(D) Private members

| Part - II

Answer the following questions

1. What is a scope?
2. Why scope should be used for variable. State the reason.
3. What is Mapping?
4. What do you mean by Namespaces?
5. How Python represents the private and protected Access specifiers?

Scope refers to accessibility of variable (2 Marks)

Based on the scope the value of the variable is taken. Ex: 24 or 25 (Global)

In the above example within function the global scope value is obtained by 'a'

Therefore the value of 'a' is 7.

| Part - III

Answer the following questions

(3 Marks)

1. Define Local scope with an example.
2. Define Global scope with an example.
3. Define Enclosed scope with an example.

4. Why access control is required?
5. Identify the scope of the variables in the following pseudo code and write its output

color:= 'Red'

mycolor():

b:='Blue'

myfavcolor():

g:='Green'

print color, b, g

myfavcolor()

print color, b

mycolor()

print color

Variable

color Global

b enclosed

g local

(Red Blue Green)

Red Blue

Red

Part - IV

Answer the following questions

1. Explain the types of scopes for variable or LEGB rule with example.
2. Write any Five Characteristics of Modules.
3. Write any five benefits in using modular programming.

(5Mar)

REFERENCES

1. Data Structures and Algorithms in Python By Michael T. Goodrich, Roberto Tamassia and Michael H. Goldwasser.
2. Data Structure and Algorithmic Thinking in Python By Narasimha Karumanchi
3. <https://www.python.org>